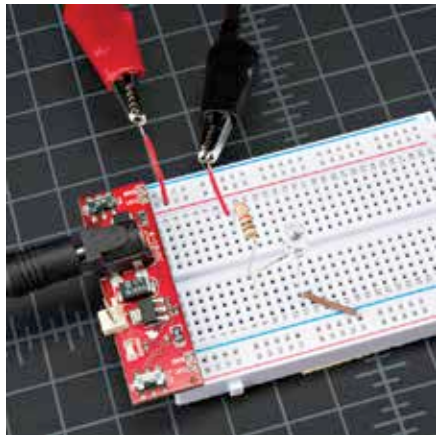


workbench and all your resistors are now completely black in colour. So how do you go about reading the resistance values now? Well, with a multimeter, of course.

To do this, pick a resistor and on the multimeter you should turn the dial to Ω or whichever is the higher value Ω . The only difference in picking the wrong Ω value is where the decimal point is going to end up, so no need to worry about blowing the multimeter. Simply, connect the probes to the two resistor legs and you should see the LCD panel start the auto-ranging process which will take barely a few seconds before it settles down to a value. This is the actual resistor value. Don't expect to see the exact value though, there will be a little variation based on the tolerance of the resistor so you might end up with some odd numbers. For example, a 10K resistor might read as 10.1K or 9.9K if it's got high tolerance or it might read exactly 10K if it's a good low tolerance resistor. If it reads 1, then change the Ω setting to a different value since you might have topped off.

Measuring current

Measuring current is extremely tricky, but it is also one of the most insightful readings in the world of electronics. The process is usually tricky since you have to measure current in series. You have to interrupt the flow of current, physically, and place the meter in-line. In our tutorial, we will use the same circuit that we used in the measuring voltage section.



You'll need an extra piece of wire to do this. Firstly, pull out the VCC wire that is going into the resistor and add your extra wire where that one was connected. Next, you will need to probe from the power pin on the power supply to the resistor. You have now broken the power to the circuit. Insert the multimeter in-line. This will now measure current as it flows through the multimeter and then into the bread board. In the picture below, alligator